

PSET-4 – Traps

Q1a

13 is stored in a2.

a0, a1, a2 holds the function arguments

```
0000000000000001c <main>:
void main(void) {
1c: 1141          addi    sp,sp,-16
1e: e406          sd      ra,8(sp)
20: e022          sd      s0,0(sp)
22: 0800          addi    s0,sp,16
printf("%d %d\n", f(8)+1, 13);
24: 4635          li      a2,13
26: 45b1          li      a1,12
28: 00001517      auipc   a0,0x1
2c: 88850513      addi    a0,a0,-1912 # 8b0 <malloc+0x106>
30: 6c6000ef      jal     6f6 <printf>
exit(0);
34: 4501          li      a0,0
36: 298000ef      jal     2ce <exit>}
```

Q1b

I believe that f and g are inline functions as I cant find them in the assembly code and the function for line 26 is working. That means that function f is being run but not called.

Q1c

```
000000000000006f6 <printf>:
```

Printf address is 000000000000006f6

Q1d

ra would contain 0x34

Q1e

Output would be He110 World

Q1f

Random/garbage value will be printed as there was only 1 argument and in printf we asked for 2.

Q2

```
AmmarKhan-Mint-OS [Running] - Oracle VirtualBox
File Machine View Input Devices Help

?:0
?:0
ubuntu-os@ubuntuos:~/xv6-labs-2025$ nano kernel/printf.c

Command 'ano' not found, did you mean:
  command 'ant' from deb ant (1.10.14-1)
  command 'anno' from deb mailutils-mh (1:3.16-1build1)
  command 'anno' from deb mmh (0.4-6)
  command 'anno' from deb nmh (1.8-1)
  command 'nano' from deb nano (7.2-2ubuntu0.1)
  command 'an' from deb an (1.2-7build2)
  command 'and' from deb and (1.2.2-4.2)
Try: sudo apt install <deb name>
ubuntu-os@ubuntuos:~/xv6-labs-2025$ nano kernel/printf.c
ubuntu-os@ubuntuos:~/xv6-labs-2025$ make qemu
mv -f fs.img fs.img.bk
riscv64-unknown-elf-gcc -Wall -Werror -Wno-unknown-attributes -O -fno-
in-strlen -fno-builtin-memset -fno-builtin-memmove -fno-builtin-memcmp
iltin-printf -fno-builtin-fprintf -fno-builtin-vprintf -I. -fno-stack-
riscv64-unknown-elf-ld -z max-page-size=4096 -T kernel/kernel.ld -o ke
nel/sysproc.o kernel/bio.o kernel/fs.o kernel/log.o kernel/sleeplock.o
/uart.o kernel/spinlock.o
riscv64-unknown-elf-ld: warning: kernel/kernel has a LOAD segment with
riscv64-unknown-elf-objdump -S kernel/kernel > kernel/kernel.asm
riscv64-unknown-elf-objdump -t kernel/kernel | sed '1,/SYMBOL TABLE/d;
mkfs/mkfs fs.img README user/_cat user/_echo user/_forktest user/_gre
ser/_dorphan user/_call user/_btest
nmeta 47 (boot, super, log blocks 31, inode blocks 13, bitmap blocks 1
ballocc: first 992 blocks have been allocated
ballocc: write bitmap block at sector 46
qemu-system-riscv64 -machine virt -bios none -kernel kernel/kernel -m

xv6 kernel is booting

hart 2 starting
hart 1 starting
init: starting sh
$ btest
backtrace:
0x80001e2c
0x80001d18
0x80001a9c
0xffffffff
$ QEMU: Terminated
ubuntu-os@ubuntuos:~/xv6-labs-2025$ addr2line -e kernel/kernel
0x80001e2c
0x80001d18
0x80001a9c
0xffffffff
/home/ubuntu-os/xv6-labs-2025/kernel/sysproc.c:72
/home/ubuntu-os/xv6-labs-2025/kernel/syscall.c:141 (discriminator 1)
/home/ubuntu-os/xv6-labs-2025/kernel/trap.c:80
?:0
ubuntu-os@ubuntuos:~/xv6-labs-2025$
```